

Krishi-Drishti: Product Roadmap & Implementation Plan

Operational Strategy Document

April 2026

1 EXECUTIVE SUMMARY

Krishi-Drishti is a verified carbon credit platform for Indian farmers. This roadmap outlines the development strategy across 4 phases (6–12 months) with clear milestones, deliverables, and resource requirements.

Timeline Overview:

- **Phase 1 (Weeks 1–8):** Core Platform & MVP
- **Phase 2 (Weeks 9–16):** Enterprise Integration & B2B
- **Phase 3 (Weeks 17–24):** Technology Enhancement (IoT, Blockchain)
- **Phase 4 (Weeks 25–32):** Geographic & Institutional Expansion

Target Launch: Month 2 (Beta with 20 farmers)

Expected Scale: 1000+ farmers by Month 12.

2 COMPETITOR COMPARISON

Krishi-Drishti is positioned as a farmer-first, India-native platform. The following table highlights the unique advantages over existing solutions.

Table 1: Competitor Analysis Matrix

Feature	Krishi-Drishti	Boomitra	EKI Energy	Grow Indigo
Verification	Satellite + ML	Satellite + Audit	Modeling + Audit	Field Audits
Payment	Direct UPI/Bank	Aggregator-based	Bank Transfer	ACH Only
Onboarding	~ 15 min	30–45 min	45–60 min	20–30 min
Language	Hindi + English	Regional	English + Hindi	English-only
Risk Detection	Real-time ML	Batch Process	Manual	Hybrid

3 IMPLEMENTATION PHASES

3.1 Phase 1: Core Platform & MVP (Weeks 1–8)

Objective: Establish the verification pipeline and launch the beta app.

- **Foundation (Weeks 1–2):** Setup AWS/GCP infrastructure, CI/CD pipelines, and PostgreSQL (PostGIS) schema.
- **Backend Core (Weeks 3–4):** JWT/OTP Authentication, KYC endpoints, and Earth Engine wrapper service.
- **Frontend MVP (Weeks 5–6):** React Native app for farmers; plot mapping and practice logging UI.
- **Carbon Engine (Weeks 7–8):** Multi-layer verification logic and ML risk scoring implementation.

3.2 Phase 2: Enterprise Integration & B2B (Weeks 9–16)

Objective: Partnership enablement and financial infrastructure.

- **Aggregator APIs:** Integration with Boomitra and EKI standardized connectors.
- **Payment Wallet:** Razorpay integration for direct farmer payouts and tax reporting.
- **Buyer Portal:** Web-based dashboard for corporate ESG buyers to purchase bulk credits.

3.3 Phase 3: Technology Enhancement (Weeks 17–24)

Objective: IoT and Blockchain for advanced transparency.

- **IoT Ingestion:** LoRaWAN/4G sensor pipelines for soil moisture tracking.
- **Blockchain:** Smart contract deployment on Polygon for credit serialization and retirement.
- **Advanced ML:** XGBoost-based fraud detection and crop classification models.

3.4 Phase 4: Expansion (Weeks 25–32)

Objective: International markets and government institutionalization.

- **Regional Expansion:** Localization for Bangladesh, Nepal, and Sri Lanka.
- **Institutional Linkage:** Integration with the Indian Ministry of Agriculture's PMKSY scheme.

4 RESOURCE ALLOCATION & BUDGET

4.1 Core Team (Monthly Estimates)

Role	Count	Estimated Cost (USD)
Backend/DevOps	4	\$3,200 – \$4,800
Frontend (Mobile)	2	\$1,200 – \$2,000
ML/Data Scientist	1	\$1,000 – \$1,500
Product/Management	1	\$1,000 – \$1,500
Total Monthly		\$6,400 – \$9,800

4.2 Infrastructure & Year 1 Totals

- **Cloud Infrastructure:** \$12,360 / year (AWS, RDS, Earth Engine).
- **Audit & Security:** \$5,000 (One-time Smart Contract audit).
- **Total Year 1 Proj.:** ~\$104,000 (Staff + Infra + Contingency).

5 RISK MANAGEMENT & GO/NO-GO GATES

Table 2: Risk Mitigation Strategy

Risk Item	Probability	Mitigation Strategy
EE API Quota Limits	Medium	Implement caching; request early quota increase.
Farmer Dropout	Medium	Focus on direct UPI payments and Hindi UX.
Regulatory Shift	Low	Modular architecture for registry compliance.

5.1 Decision Gates

- **Gate 1 (Week 8):** MVP must onboard 20 farmers with < 7 day verification speed.
- **Gate 2 (Week 16):** Payment system must process \$10k/month with zero failures.
- **Gate 3 (Week 24):** ML models must reach > 99% accuracy in fraud detection.

6 CONCLUSION

The Krishi-Drishti roadmap ensures a rapid time-to-market (8 weeks) while building the institutional and technological depth (Blockchain, IoT) required for long-term scalability in the global climate finance ecosystem.